

W05 Superposition of Waves

By the end of this topic, you should be able to:

7. Vibrations and waves

7.3 Superposition of waves

7.3.1 The principle of superposition and the formation of stationary (standing) waves.

Candidates should be able to:

a) State the principle of superposition of pulses and waves.	
b) State the conditions for stationary waves. It should be assumed that the stationary waves are planar and complete cancellation occurs at the nodes.	
c) Identify the formation of standing waves in a given system. Examinable situations will involve either two generators of waves or a generator of waves and reflector. Questions involving open ends will not be set.	
d) Use displacement-position graphs to explain how nodes and antinodes are formed.	
e) Define nodes and antinodes.	
f) Identify nodes, antinodes and loops in a given standing wave.	
g) Identify and calculate the wavelength in a stationary wave.	
h) Distinguish between progressive and standing waves in terms of the amplitude, phase relationship, the propagation of the wave profile and energy as applied to a sinusoidal wave.	

7.3.2 Standing waves on a string.

Candidates should be able to:

a) Identify vibrating strings as a system that can resonate at various frequencies.	
b) Describe in detail Melde's experiment or a similar experiment involving an oscillator to demonstrate the formation of stationary waves in strings and to investigate the relationship between the frequency f_n of the n th harmonic, the length L of the string that is free to vibrate, the tension force T in the string and the mass per unit length μ of the string. This includes providing a diagram, the procedure, observations, and measurements made.	
c) Describe qualitatively how a standing wave can be formed on a stretched string.	
d) State and use the equations $f_n = \frac{n}{2L} \sqrt{\frac{T}{\mu}}$ and $v = \sqrt{\frac{T}{\mu}}$ where v is the speed.	

7.3.3 Diffraction of water waves at a gap

Candidates should be able to:

a) Describe an experiment to investigate the diffraction of water waves at a gap. This includes providing a diagram and observations made.	
b) Explain and sketch how the diffraction pattern would change as the width of the gap and the wavelength are varied.	

7.3.4 Diffraction of visible light and microwaves at a single slit (rectangular aperture).

Candidates should be able to:

a) Describe an experiment to investigate the diffraction of microwaves at a single slit. This includes providing a diagram and observations made.	
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b) Describe an experiment to investigate the diffraction of visible light at a single slit. This includes providing a diagram and observations made.	
c) Explain and sketch how the pattern would change as the width of a rectangular aperture and the wavelengths are varied.	
d) State and use the equation $\sin(\theta) \approx \theta = \frac{\lambda}{a}$ to solve problems as well as to describe and explain the behaviour of waves at a single slit, where a is the width of a rectangular aperture, λ is the wavelength and θ is the angular distance between the normal and the first minimum that is measured in radians.	
e) Sketch the intensity graph for a single rectangular aperture.	

7.3.5 Resolving power of instruments

Candidates should be able to:

a) Discuss how the diameter of a circular aperture and the wavelength of the incident wave affect the resolving power of an instrument.	
b) State and use the equation $\sin(\theta) \approx \theta = 1.22 \frac{\lambda}{a}$ where a is the diameter and θ is the angle between the normal and the first diffraction minimum.	
c) Explain what is meant by resolving power of an instrument.	
d) State and use Rayleigh's criterion for an instrument with a circular aperture.	
e) Explain how the resolving power of an instrument with a circular aperture changes with wavelength and diameter.	

7.3.6 Interference of waves from two coherent point sources

Candidates should be able to:

a) Explain what is meant by two coherent sources.	
b) Sketch the resulting wavefronts to describe qualitatively the regions of constructive and destructive interference of waves originating from two coherent point sources.	
c) Explain what is meant by path difference and express the path difference as a multiple of the wavelength.	
d) Determine the phase difference at a point taking into account the path difference, the initial phase relationship between the sources and any other phase change.	
e) State and use the conditions for constructive and destructive interference in terms of the phase difference.	
f) Describe how the interference pattern arising from waves emitted from two point sources relates to the phase differences.	
g) Describe qualitatively how the interference pattern would change if the source separation and the wavelength are changed.	

7.3.7 Young's double-slit experiment using visible light

Candidates should be able to:

a) Define monochromatic light.	
b) Describe in detail Young's double-slit experiment for visible light and use it as experimental evidence of the wave nature of light. This includes providing a diagram, the procedure and observations made.	

c) State typical length scales that are used in Young's double-slit experiment.	
d) Sketch the interference pattern (fringes) observed on a screen in Young's double slit experiment.	
e) Identify the central (zeroth order) bright fringe and the n^{th} order bright (dark) fringes.	
f) State the conditions for the interference pattern to be observable.	
g) State how the pattern would change if the slit separation, wavelength and distance between the slits and the screen are changed.	
h) Sketch the intensity graph for a double-slit.	
i) State and use the equation $y = \frac{\lambda D}{d}$, where y is the fringe separation, λ is the wavelength, d is the slit separation and D is the distance between the double-slits and the screen.	

7.3.8 Optical diffraction grating

Candidates should be able to:

a) Describe what a diffraction grating is.	
b) State and use the equation $d = \frac{1}{N}$ where d is the grating spacing and N is the number of lines per unit length.	
c) State and use the equation $d \sin(\theta_n) = n\lambda$, where θ_n is the angular separation between the normal and the n^{th} order spectral line (image).	
d) Explain how the pattern changes when the wavelength is changed and when more than one wavelength (including white light) is used.	
e) Compare the spectra produced by a diffraction grating with that produced by a prism.	
f) Describe in detail an experiment involving a spectrometer and a diffraction grating to measure the wavelength. This includes providing a diagram, the procedure, the measurements and the calculations made. Details of the adjustments to the spectrometer will not be examined.	

5.1 Stuff You Should Already Know

- ☛ Displacement is a vector quantity; it has both magnitude and direction.
- ☛ The displacement y of a particle undergoing simple harmonic motion (SHM) varies with time t according to $y = y_0 \sin(\omega t \pm \phi)$. This equation also applies to progressive waves.

$$\omega = \frac{2\pi}{T} = 2\pi f \text{ and } \phi = \frac{2\pi x}{\lambda}$$
- ☛ The speed of a mechanical wave depends on the medium through which it is travelling.
- ☛ Resonance occurs when the forcing frequency equals the natural frequency. During resonance, a large amplitude is produced and there is a maximum transfer of energy.
- ☛ If $a \propto b$, then $a = kb$
- ☛ $2\pi \text{ rad} = 360^\circ$

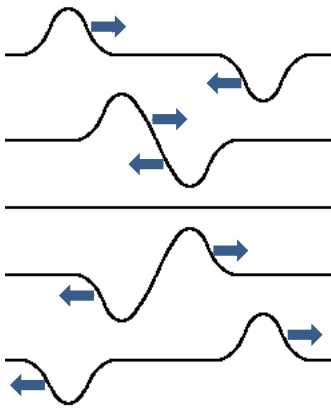
5.2 The Principle of Superposition

When two pulses or waves of the same type meet each other, their **amplitudes add up** to produce the shape of the resulting pulse or wave.

This is known as the **principle of superposition** →

The resultant displacement at any point is the **sum** of the separate displacements due to the two waves

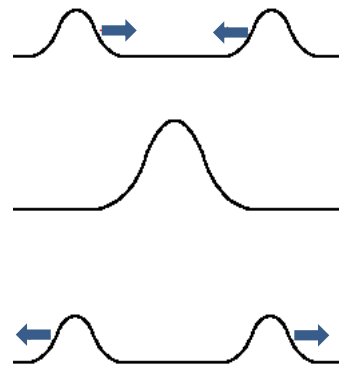
Two identical pulses are travelling on a string (diagrams).



← When they point in opposite directions, the total amplitude is zero and the string flattens for one instant. This is **destructive interference**.

When the pulses point in the same direction, the total amplitude is doubled. This is **constructive interference**. →

The pulses then pass through each other.



E.g. 5-1: The two transverse pulses shown are approaching each other.



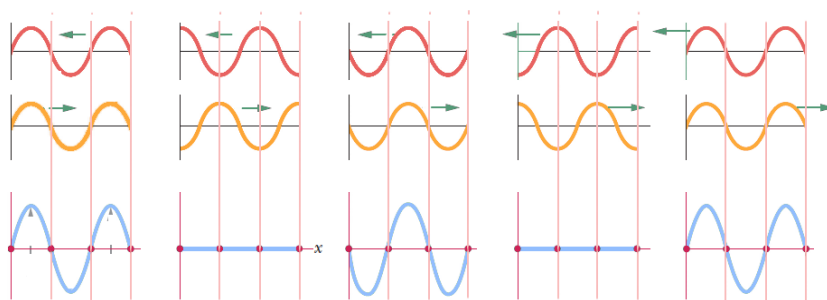
- Draw the resultant displacement when the two pulses meet.
- State the type of interference obtained.

CW 5-1: MS Forms Quiz

5.3 Stationary (or Standing) Waves

The term *stationary* or *standing* refers to two progressive waves which meet and form a wave that appears to be stuck in place.

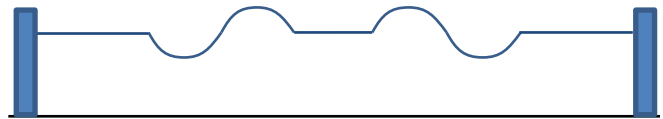
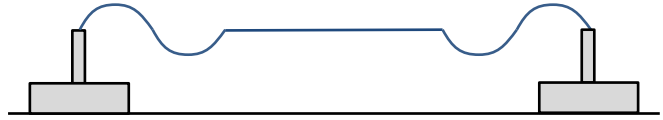
Standing waves form when two waves with equal amplitude and frequency travelling at the same speed in opposite directions superimpose.



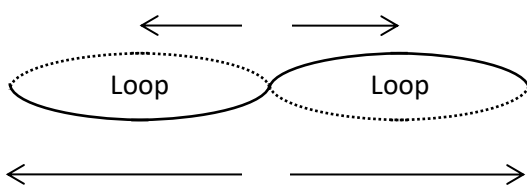
Almost all the particles in standing waves oscillate with SHM, but with many different amplitudes. A few particles do not oscillate at all.

Standing waves can be set up between:

- Two wave sources (generators) set some distance apart, which have the same frequency, amplitude and vibrating in the same medium (thus ensuring the same wave speeds).
 - This will not work for light waves because any small variation in the frequency of either light source will collapse the standing wave. To overcome this, we take one wave source and split it into two, and then aim the split parts at each other. Since these split parts are moving in opposite directions with the same speed, frequency and wavelength, then they form standing waves when they superimpose.
- A wave generator close to a reflecting surface, so that there will be little damping. The reflected waves will have the same frequency, speed and amplitude of the waves coming from the source.
- The ends of a taut string like in stringed instruments. When a string is plucked, two waves will travel to the edges and then reflect along the string. The reflected waves have the same speed, frequency and amplitude, and will thus form a standing wave when they meet.



The diagram below shows a string fixed at both ends vibrating up and down.



← The non-moving points in the wave are called nodes (N). The other particles in the wave have different maximum displacements i.e. different amplitudes. The greatest amplitudes are called antinodes (A). Each loop is equivalent to half a

standing wave, thus the distance between two adjacent nodes (or antinodes) is equal to half a wavelength. The wavelength λ of a standing wave is equivalent to the length of two loops.

Nodes form where points in the two waves which are being superimposed always have equal displacements but in opposite directions, i.e. the points are in antiphase, so they always cancel each other out.

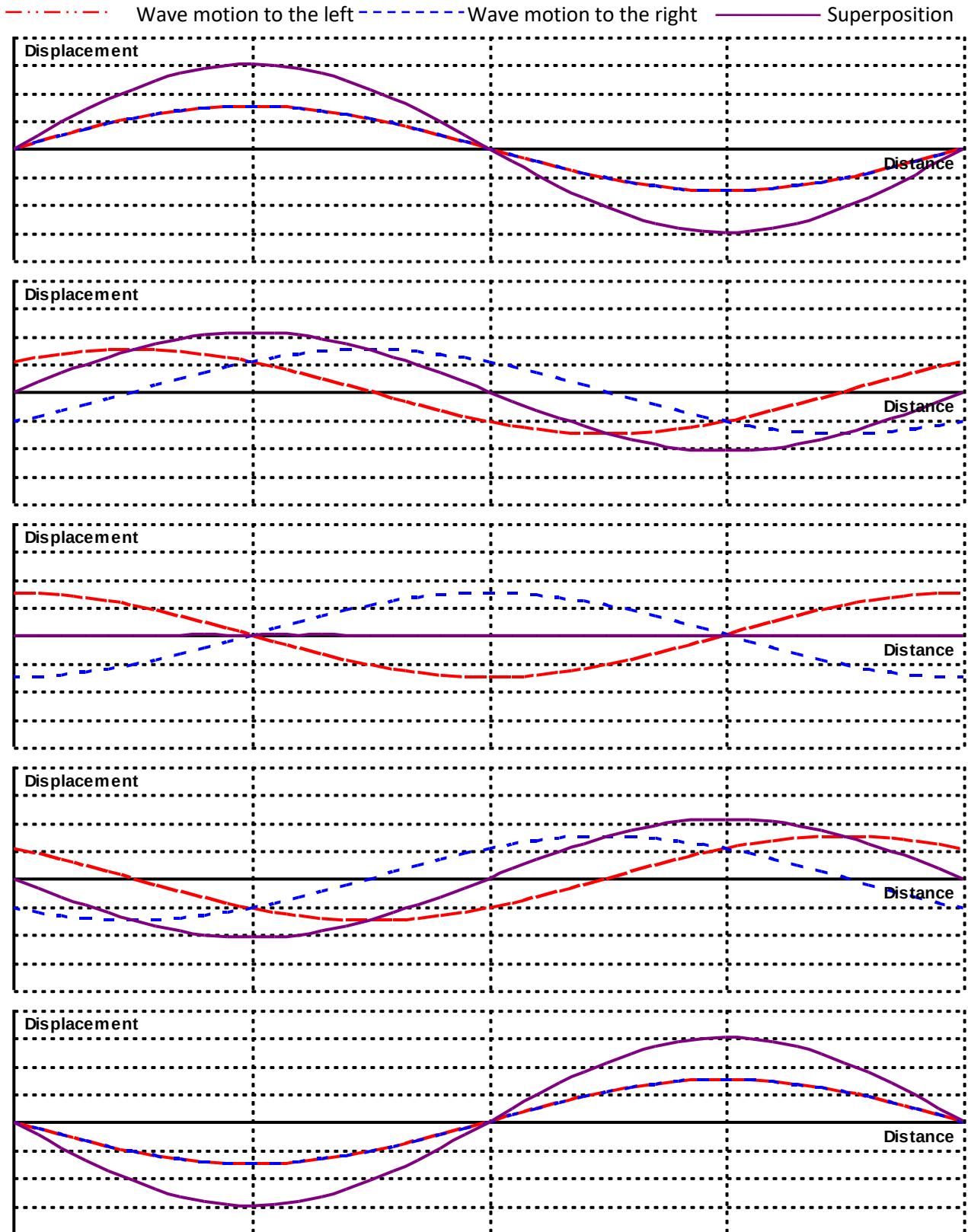
Exactly half-way between two adjacent nodes we can find an antinode, where points in the two progressive waves which make up the standing wave are exactly in phase, so they form constructive interference.

A **node** is a point along a standing wave where the wave has minimum (zero) amplitude.

An **antinode** is a point along a standing wave where the wave has maximum amplitude.

Note: All the particles in progressive waves go through equal oscillations. Standing waves have non-moving particles and various displacements.

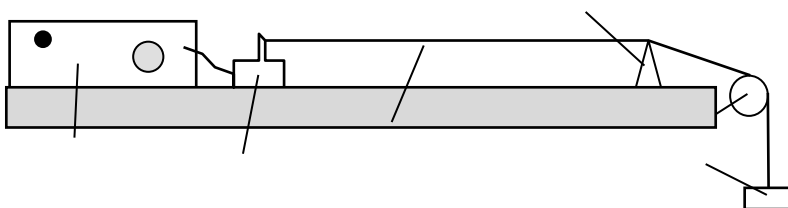
The following graphs explain the formation of standing waves through half a cycle.

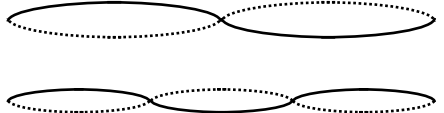


The two waves (dashed lines) above are travelling in opposite directions at the same speed. They have the same amplitude. Their superposition (solid line) forms the standing wave. The points where the superposition curves intersect the x-axes are the nodes of the standing waves.

5.4 Standing Waves in Strings: Harmonics

1. Set up the apparatus as shown.



2. Pluck the string in the *middle*. It will vibrate at its natural frequency (f_1).
3. Switch on the signal generator and *slowly* increase its frequency. At first, the string's amplitude will be *small*. When the forcing frequency *equals* the natural frequency, *resonance* occurs. The amplitude will become *large* and a *standing wave* with one *loop* forms. The frequency f_1 is called the *fundamental frequency*, or the first harmonic.
4. Slowly, continue to *increase* the vibrator frequency. The amplitude will be *small* again until another *frequency* (f_2) causes resonance. This happens when $f_2 =$  f_1 . The standing wave now has *two* loops. This new frequency is called the second harmonic.
5. Increase the frequency until the third harmonic $f_3 =$  f_1 . *Resonance* occurs again and the *standing wave* forms with *three* loops.
6. Obtain standing waves with four and five loops by increasing the forcing frequency to the fourth and fifth harmonics respectively.

In general, for a harmonic n :

for a vibrating string of length L :

5.5 Progressive and Stationary Waves: Properties and Comparisons

Transverse Progressive Waves	Transverse Stationary Waves
Carry energy	Trap energy
All particles vibrate in SHM - there are no nodes	There are points, called nodes, where the displacement is always zero
Within a half wavelength, some particles are going up, the rest are going down – none of the particles are in phase	Within a loop, all particles oscillate in phase - they are all going up or going down. The oscillations in one loop are in antiphase with those in an adjacent loop.
At any instant, a particle within a half wavelength is at the maximum displacement. All particles reach the amplitude at one time or another.	The particles within a loop reach their maximum displacement simultaneously. The greatest amplitude is called the antinode.

E.g. 5-2: An 80 cm string is set vibrating at a frequency of 50 Hz, forming a standing wave with one loop. Calculate the wavelength and the speed of the standing wave.

E.g. 5-3: A standing wave with five loops is set up in a string when the forcing frequency was 150 Hz.

a) Calculate the fundamental frequency.

b) If the wavelength is 24 cm, calculate the length of the string.

E.g. 5-4: A vibrator connected to a string vibrates at 150 Hz and produces a standing wave with three loops. Calculate the frequency of the second harmonic. How many loops will this standing wave have?

CW 5-2: **MS Forms Quiz**

CW 5-3: **MS Forms Quiz**

HW 5-1: When four loops are observed in a standing wave, the forcing frequency is equal to 200 Hz. The length of the stretched string is 80.0 cm.

- a) How do standing waves form? (2)
- b) Calculate the natural frequency of the stretched string (1)
- c) Find the distance between two antinodes and hence the wavelength (2)
- d) Calculate the speed of the standing waves (2)
- e) Name three differences between standing and progressive waves (3).

5.6 *The Laws of Vibrations of Stretched Strings (Including Melde's Experiment)*

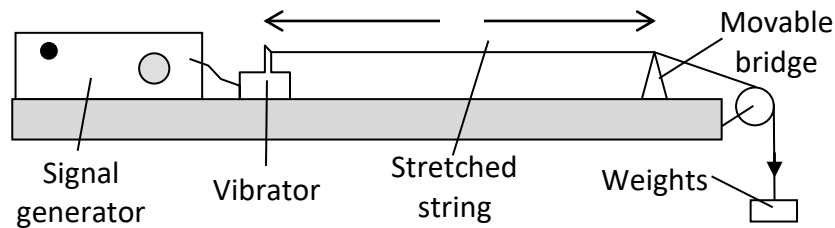
Guitarists must do at least one of the following to obtain these different frequencies:

- Press a string with a finger before plucking it - now a shorter string vibrates.
- Turn the tuning peg of the string to change the tension in the string.
- Pluck a thinner or a thicker cord.

The following experiments are designed to investigate which factors affect the frequency of a standing wave in a vibrating string fixed at both ends. There are variations on these experiments, Melde's version being one of them.



Set up the apparatus below to investigate how frequency varies with length, tension and mass per unit length of stretched strings.



There are five quantities that can be varied using this apparatus. In the following experiments, we need to investigate how the frequency varies as one other quantity is changed. The other quantities that are kept constant are called

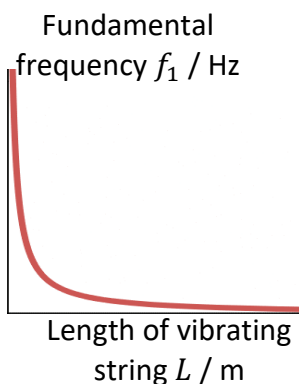
- Frequency f may be determined from the dial on the signal generator, but may be measured more accurately using a cathode ray oscilloscope.
- Length L of the string may be measured with a metre ruler or tape measure.
- Tension T in the string is equal to magnitude of the hanging weight.
- Mass per unit length μ is obtained by measuring the string's mass and length and dividing.
- Harmonic n which is equivalent to the number of loops seen in the string when a standing wave is set up.

The following experiments are described as being carried out at the first harmonic, or fundamental frequency, f_1 . This means that $n = 1$, but each experiment can be performed at higher harmonics.

5.6.1 Relationship between Frequency and Length of Vibrating String

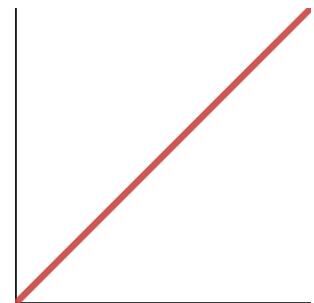
Using the same string and same hanging weight throughout, thus

Measure the length L of the string stretched between the vibrator and the bridge. Switch on the vibrator and slowly increase its frequency with the signal generator until resonance occurs and you get a standing wave with one loop. Take note of this fundamental frequency f_1 . Repeat for other lengths by moving the bridge.



← Plotting a graph of fundamental frequency against the length of vibrating string gives a curve which shows that as the length of vibrating string decreases, the fundamental frequency increases.

A graph of fundamental frequency against the reciprocal of the length of vibrating string gives a straight line through the origin. →



Conclusion:

Fundamental frequency is

to the length of a vibrating string

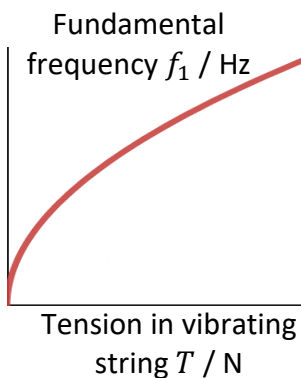
if the tension in the string is kept constant.

5.6.2 Relationship between Frequency and Tension in Vibrating String

Keeping the same length of vibrating string throughout, the control variables are

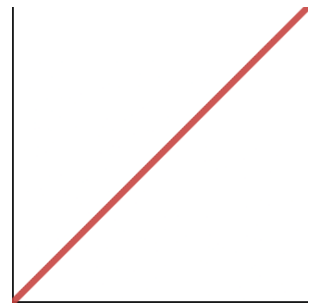
This experiment can be extended to determine the magnitude of the mass per unit length μ of the vibrating string. This is known as **Melde's experiment** and this extension is described in the important note below.

Take note of the hanging weight, which is equal to the tension T / N in the string. Pluck the string in the middle to set it oscillating at its natural frequency. Slowly, increase the vibrator's frequency from the signal generator until resonance occurs and you get a large amplitude. Take note of this frequency. Repeat for other weights.



← Plotting a graph of fundamental frequency against the tension in the vibrating string gives a curve which shows that as the tension increases, the fundamental frequency increases at an ever-lower rate.

A graph of the square of fundamental frequency against tension in the string gives a straight line through the origin. →



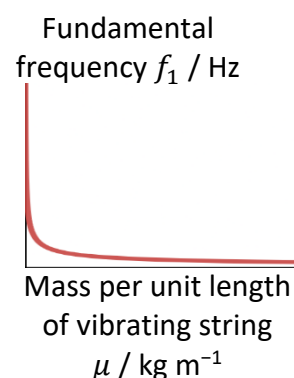
Conclusion:

Fundamental frequency is proportional to the square root of the tension in the vibrating string if the length of the string is kept constant.

5.6.3 Relationship between Frequency and Mass per Unit Length of the String

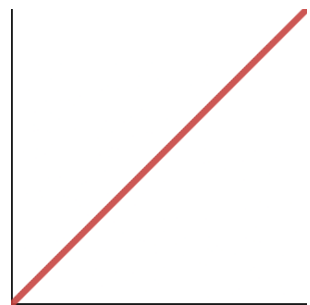
Now we will use strings of different diameters and/or densities, i.e. strings having different masses per unit length μ / kg m⁻¹, while always keeping the same vibrating length and the same hanging weight, thus the control variables are

Slowly, increase the vibrator's frequency from the signal generator until resonance occurs and you get a large amplitude. Take note of this frequency. Change the string and repeat.



← Plotting a graph of fundamental frequency against the mass per length of vibrating strings gives a curve which shows their fundamental frequency increases with thickness.

A graph of the square of fundamental frequency against the reciprocal of the mass per unit length of the vibrating string gives a straight line through the origin. →



Conclusion:

Fundamental frequency is proportional to the square root of mass per unit length of vibrating strings if the strings are all of equal lengths and have equal tension.

The above experiments can be repeated for frequencies of any harmonic f_n , where $n = 1, 2, 3, \dots$, and similar results and conclusions will be obtained.

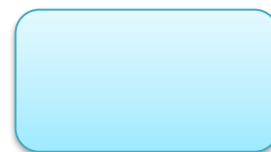
Summary: For any harmonic frequency f_n of the stretched string be, its vibrating length L , tension T and mass per unit length μ vary like this $\rightarrow$

if T and μ are constant
if L and μ are constant
if L and T are constant

Combining the results: $f_n \propto \frac{1}{L} \sqrt{\frac{T}{\mu}}$.

Since and for any wave in a given medium: $f \propto \frac{1}{\lambda} = \frac{n}{2L}$, then:

This equation is known as the **wave equation for a stretched string** $\rightarrow$



Density $\rho = \frac{m}{V} =$ and $\mu = \frac{m}{L} \therefore \rho =$ Or $\therefore$

Also, since $v = f\lambda$,
then $\rightarrow$



or

Important Note: If you are asked to describe Melde's experiment, then you need to describe the experiment of how frequency varies against tension, and then add the following:

Comparing the **wave equation for a stretched string** to the general equation for a straight line, we can see that the gradient m of the graph is:

$$f_n = \frac{n}{2L\sqrt{\mu}} \sqrt{T}$$

$$y = m x + c$$

$m =$ $\therefore \mu =$

Thus, you need to measure the length L of the vibrating string and calculate the gradient of the graph. Then plug these values, and the harmonic mode used in the experiment, to determine the mass per of the vibrating string.

Next, measure the mass of the string and its total length and divide the two values, to find a direct value of μ , and compare the two values.

E.g. 5-5: A guitarist wants to increase the frequency of a note by 1.5 times. If the guitar string is originally 90 cm long, how long should the vibrating string be for the new frequency to be heard?

E.g. 5-6: A string has twice the tension of another identical string. The natural frequency of the taut string is 256 Hz. Find the natural frequency of the looser string.

E.g. 5-7: Every octave in a piano means that the frequency is doubled. Compare the mass per unit length of two strings in a piano, which are an octave apart, if they have the same tension and length.

E.g. 5-8: A 75 cm long string, of mass 6.0 g, is under a tension of 120 N. Calculate:

a) The mass per unit length of the string

b) The tension needed to double the natural frequency of the string.

c) The frequency of a standing wave with 5 loops that forms in the string.

CW 5-4: A 64 cm long string, of mass 10 g, is under a tension of 400 N. Calculate:

a) The frequency in the string if it has a standing wave with two loops. [250 Hz]

b) The frequency of the first harmonic.

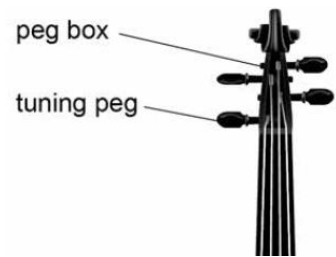
c) Draw diagrams of the two standing waves, which should compare the relative lengths and amplitudes.

Name and label at least one point which does not vibrate.

Name and label another point that has a maximum amplitude.

CW 5-5: **MS Forms Quiz**

CW 5-6: The figure on the right shows the top part of a violin. The tension in the strings can be increased or decreased by rotating the tuning pegs.



- a) The vibrating length of a violin string is 0.33 m.
- i) When the tension in a string is 25 N, the string vibrates with a first-harmonic frequency of 370 Hz. Calculate the string's mass per unit length. [$4.19 \times 10^{-4} \text{ kg m}^{-1}$]

- ii) Determine the speed at which waves travel along the string. [244 m s^{-1}]

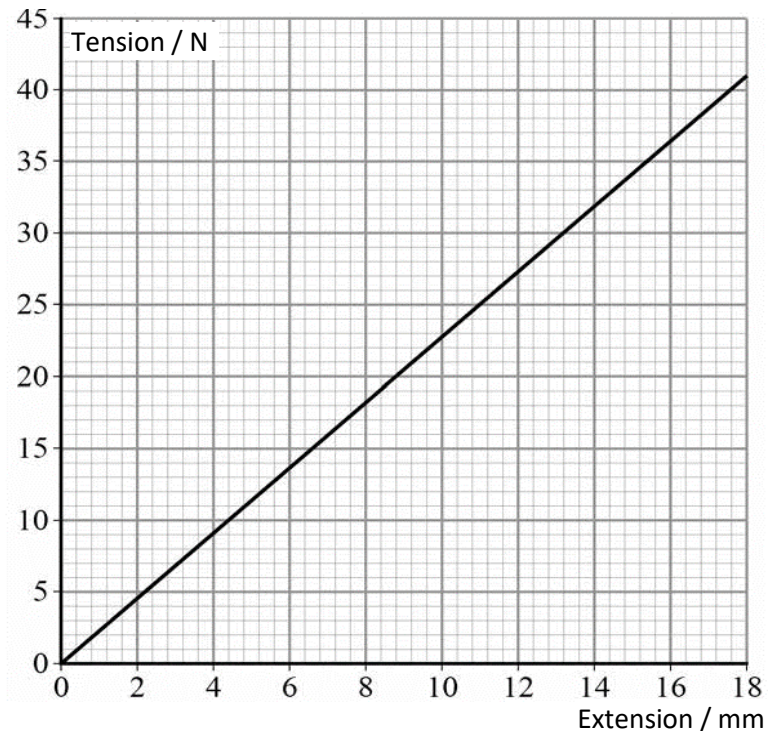
- b) The graph (right) shows how the tension in the string varies with the extension of the string.

The string with its initial tension of 25 N is vibrating at a frequency of 370 Hz

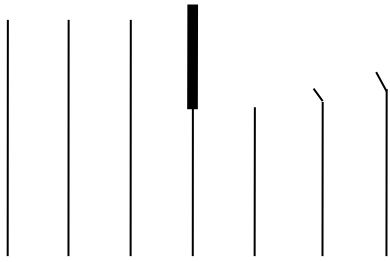
The diameter of the circular peg is 7.02 mm.

Determine the higher frequency that is produced when the string is stretched by rotating the tuning peg through an angle of 75°

Assume that there is no change in the diameter of the string. [438 to 444 Hz]

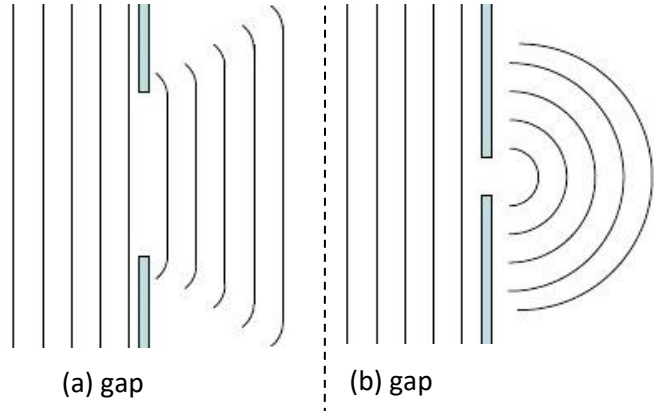


5.7 Diffraction



Diffraction is a property of any wave motion. Any wave spreads after passing the edge of a solid object (left) or after passing through a narrow space (right), instead of continuing to travel in a straight line. As the waves spread, they tend to cancel out or reinforce each other in different directions, thus causing characteristic rippled patterns of waves.

If the size of the gap or slit through which the waves pass is large when compared to the wavelength, the waves do not spread much (a). When the size of the gap is almost equal to the wavelength, the waves are spread out more (b). If the gap is smaller than the wavelength, the wave cannot pass through and is reflected back.



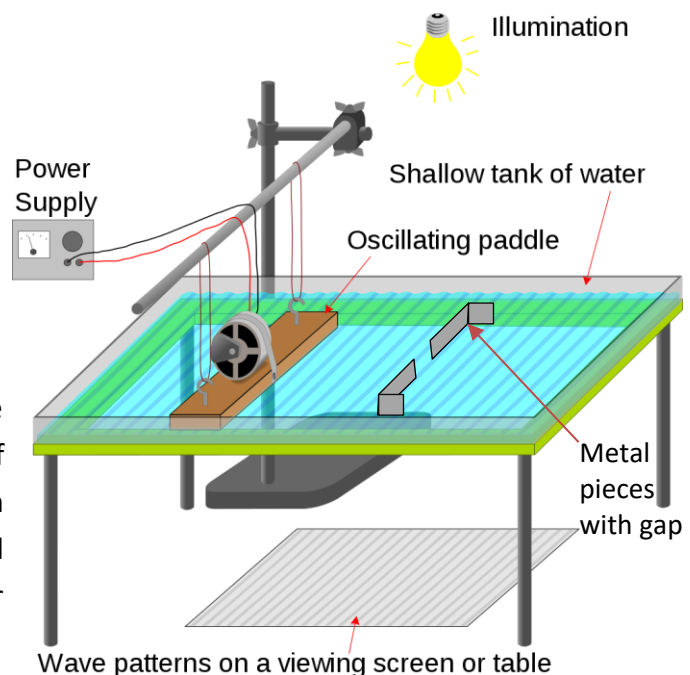
All progressive waves, transverse or longitudinal, can be diffracted when they pass an obstruction or through a gap.

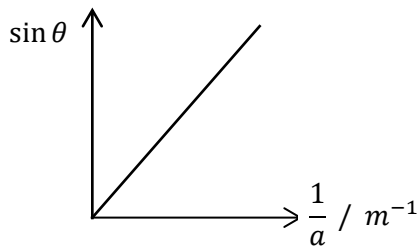
In optical instruments, the spreading out and blurring of light by diffraction limits magnification and sharpness of the image. Details smaller than 10^{-8} m cannot be seen by most optical microscopes, so in order to observe them, one must use waves of a wavelength which is smaller than that of light.

5.7.1 Experiment: Diffraction of Water Waves

Apparatus: Ripple tank, lamp, motor, wooden bar, power supply, white paper, two pieces metal bar, ruler, protractor and stroboscope.

Method: Set up the apparatus as shown. Use the metre ruler to measure the width a of the gap between the metal pieces. Switch on the power supply to produce plane waves which propagate towards the aperture (gap). The motion of the waves can be “frozen” by switching on the stroboscope. Use the protractor to find the angle θ that the waves bend. Repeat for 5 different values of the gap size a .



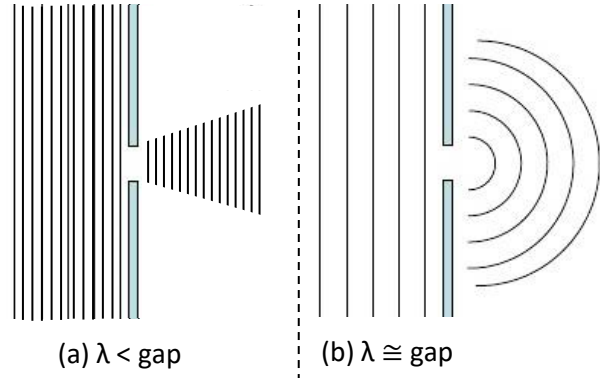


Plot a graph of $\sin \theta$ against $\frac{1}{a}$. Since the graph is a straight line through the origin, then $\sin \theta$ is inversely proportional to the width of the gap.

$$\sin \theta \propto \frac{1}{a} \therefore$$

During the experiment, the wavelength λ was kept constant.

The experiment can be repeated by varying the frequency of the oscillating paddle while keeping the gap at a constant size. At high frequency, the wavelength is small and the waves do not spread much after the gap. As the frequency is decreased, the wavelength increases and the waves spread more ($\sin \theta$ increases) because the wavelength approaches the size of the gap.



A graph of wavelength against $\sin \theta$ will give a straight line through the origin, which means that they are directly proportional.

$$\sin \theta \propto \lambda \therefore$$

Combining the results of the two experiments:

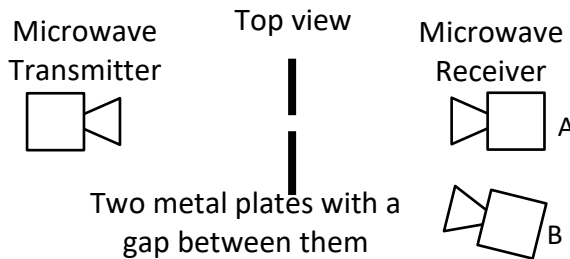
5.7.2 Experiment: Diffraction of Microwaves at a Single Slit

Apparatus:

Microwave transmitter, two metal plates, microwave receiver (detector)

Diagram:

Method:



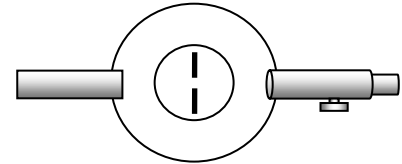
1. Set up the apparatus as shown in the diagram.
2. Switch on the microwave source.
3. Place the microwave receiver in the path of the central maximum (position A). This is the point where the waves pass through the gap without undergoing diffraction.
4. Move the receiver to one side of the central maximum while keeping the same distance from the slit until microwaves are not detected anymore (position B). Record the receiver's position.
5. Repeat Step 4 but on the other side of the central maximum, maintaining the same distance from the slit.
6. Repeat with different gap widths.

Observations:

1. As you move away from the central maximum towards the sides, the intensity of the received microwaves gradually decreases. This is indicative of diffraction occurring at the slit.
2. The wider is the width of the slit, the less the microwaves spread after passing the slit.

5.7.3 Experiment: Diffraction of Light at a Single Slit

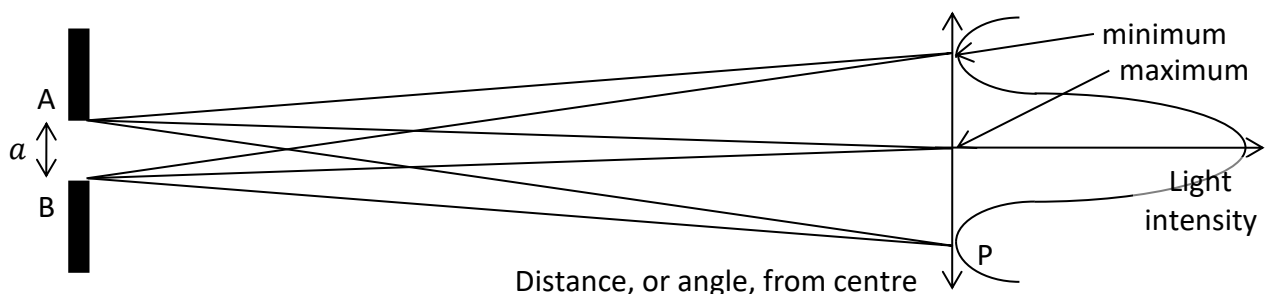
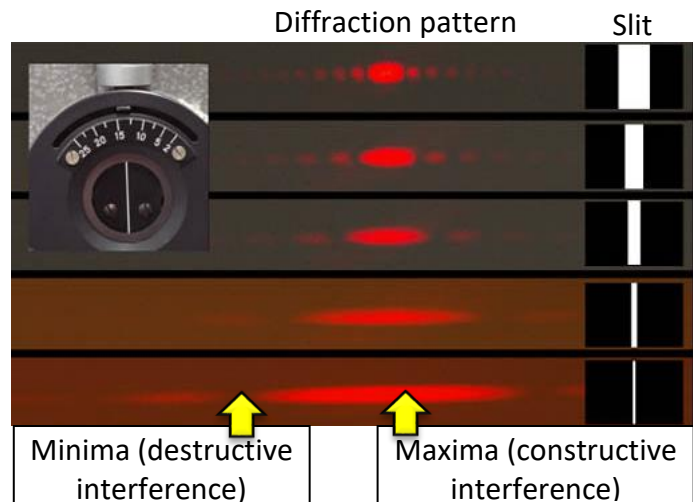
An adjustable slit is placed on the table of a spectroscope. A monochromatic (**single colour**) LASER is viewed through it using



the spectroscope telescope.

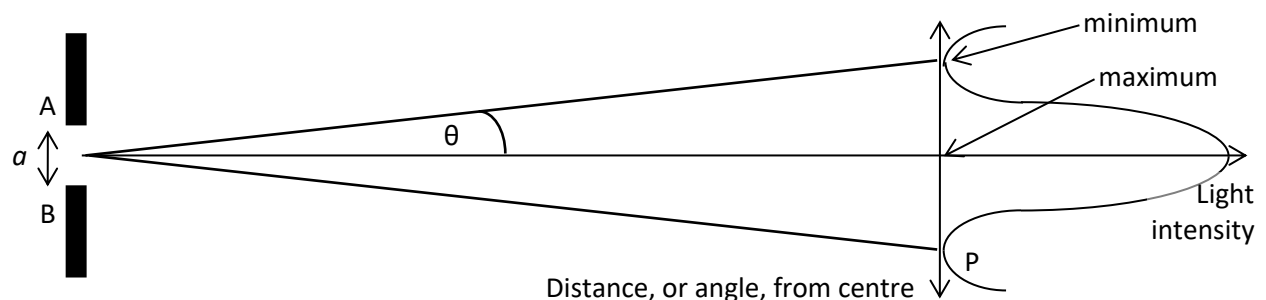
As the slit width is narrowed, the diffraction pattern broadens. The pattern disappears when the slit width is smaller than one wavelength of the light used.

These patterns occur because at the centre, where the bright fringe forms, waves from the slit arrive in phase resulting in constructive interference. Away from the centre, waves arrive out of phase, so light is fainter. At some point P, the wave leaving from edge A of the slit is half a wavelength longer than that leaving edge B, resulting in destructive interference. This is the centre of the first dark fringe.



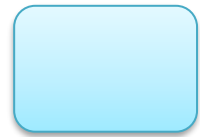
Note: in the above figure, the slit width as drawn is too large when compared to the practical width and that $(AP - BP)$ is of the order of 10^{-5}m , i.e. $AP \approx BP$.

Although the difference in length between the light rays AP and BP is enough to cause interference, as far as angles are concerned, $AP = PB$. So, the above diagram can be simplified as shown below, by considering AB as a point.



Let the aperture (slit) width $AB = a$ and let the angle subtended by the centre of the bright region and the centre of the dark region be θ .

Then $\rightarrow$



To **broaden** the diffraction pattern (i.e. to increase θ), either **decrease** the aperture **width** or use light of a **longer wavelength**.

Note: for very small angles in radians, $\sin \theta \approx \theta$. Mathematically, as $\theta \rightarrow 0$, $\sin \theta \rightarrow \theta$. Hence, $\sin \theta = \frac{\lambda}{a}$ and $\theta = \frac{\lambda}{a}$ are equivalent if θ is in radians and very small (smaller than 6° or $\frac{\pi}{30}$ rad).

5.8 Resolving Power of Optical Instruments

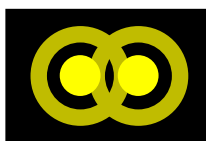
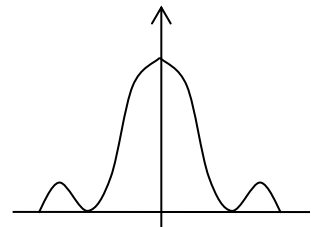
The resolving power θ of an optical instrument, such as your eye, or a telescope, is its ability to separate far-away objects that are close together into individual images, as opposed to a single merged image. If you look at two stars in the sky, for example, you can tell they are two stars if they're separated by a large enough angle. Some stars, however, are so close together that they look like one star. You can only see that they are two stars by looking at them through a telescope. So, why does the telescope resolve the stars into separate objects while your eye cannot? It's all due to diffraction.

Resolving Power is the ability of an optical instrument to form separate images of objects that are very close together.



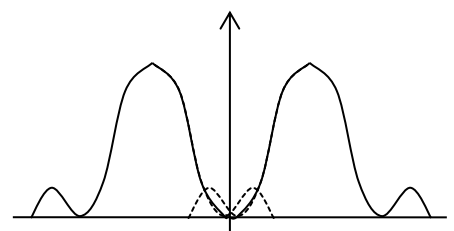
$\leftarrow$ Such a diffraction pattern forms on your retina when you look at a far-away point object. The central white disc is the central bright maximum. The black circle is the first minimum. The circle is the next bright fringe.

This is how light intensity varies with distance (or angle) $\rightarrow$



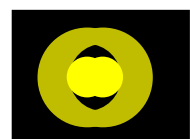
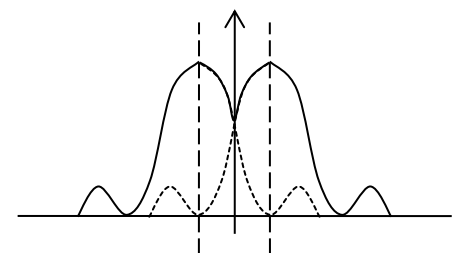
$\leftarrow$ For two objects separated by a small angle, still viewed through a circular aperture, the diffraction patterns will overlap. You are able to resolve the two objects as long as the central peaks in the two diffraction patterns do not overlap.

This is how light intensity varies with distance (or angle) $\rightarrow$



$\leftarrow$ The limit occurs when one central peak falls at the position of the first dark fringe for the second diffraction pattern. At this position, the two objects are **just resolved**.

This is known as **Rayleigh's criterion**.



$\leftarrow$ If the two central peaks overlap, the two objects will look like one. They will be indistinguishable from each other. The intensity-distance graph will have a similar shape as that for a single source.

The resolving power does not necessarily increase when the magnification is increased; if a blurred image is simply magnified, the result is a larger blurred image. The factors that affect the resolving power of an instrument are the size of the aperture, which can be a slit or a circular hole, through which objects are viewed, and the wavelength of the light used.

Imaging instruments have a circular aperture, not a slit. This difference gives rise to a multiplying factor of 1.22 in the formula for diffraction by a single slit.

When viewing images through a circular aperture, i.e. optical instruments and your eyes, the equation must be rearranged →

Consider two objects, a distances s apart and D from the optical instrument. They are just resolved. The relation between resolving power θ and these two lengths (s and D) is



For small θ in rad,
 $\tan \theta \approx \theta \approx \sin \theta$

Rectangular apertures

Circular apertures

For imaging instruments, the resolving power is improved by increasing the diameters of the apertures and by using shorter wavelengths. When using coloured filters, resolution is better using blue light than red, because its wavelength is smaller.

The wavelengths of radio waves are much larger than those of light, so radio telescopes must be huge to compensate.

A 76 m (diameter) radio telescope at Jodrell Bank in Cheshire, UK. →



E.g. 5-9: Laser light, of wavelength 448 nm, is aimed at a single slit. The angle between the central bright band and the centre of the first dark fringe is $\frac{\pi}{3600}$ rad. The distance between the slit and the screen is 1.20 m. Calculate:

a) The slit width.

b) The width of the central bright fringe.

c) The distance between the centre of the brightest fringe and the centre of the first dark fringe when light of wavelength 668 nm is aimed at the slit.

d) The wavelength, in nm, of an infra-red beam passed through the slit when a central bright fringe forms which is 3.52 mm wide.

E.g. 5-10: Estimate the size of lunar craters that can be just resolved using a telescope fitted with a lens of diameter 100 mm, given that the wavelength of light used is 600 nm. The distance from the Earth to the Moon is 380 000 km.

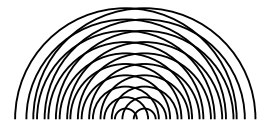
CW 5-7: [MS Forms Quiz](#)

CW 5-8: [MS Forms Quiz](#)

HW 5-3: [MS Forms Quiz](#)

5.9 Interference

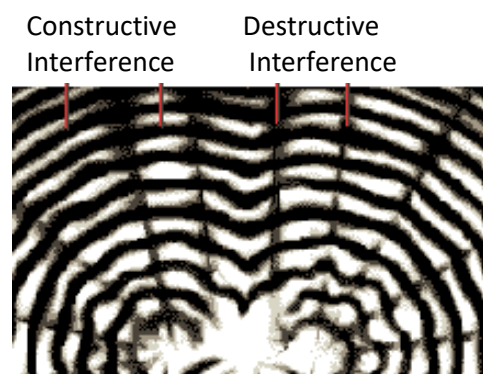
If two loudspeakers are about 0.5 m apart and connected to the **same** signal generator producing an output frequency of around 1 kHz, then at different positions different intensities of sound are heard. The waves spread out in a circular pattern from each loudspeaker. In places where two compressions meet we get constructive interference (by the principle of superposition). The resultant amplitude will be twice that produced by a single loudspeaker. At these places you will be able to hear a louder note. At other places, a compression meets a rarefaction. By the principle of superposition, the resultant amplitude will be zero, due to destructive interference. At these places, in theory, you will not hear any sound. In practice, you will hear a note with a low volume since most of us have two ears! One ear would be in a 'no noise' point while the other in a reduced noise point.



Since both loudspeakers are connected to the same signal generator, they produce **coherent** waves.

Coherence: two waves with the *same frequency* (or wavelength) and a *constant phase difference*.

Interference patterns can be produced with water waves on the surface of a ripple tank. Two dippers, attached a few centimetres apart on a horizontal beam, are made to vibrate. Each dipper will be the source of circular waves. The diagram shows the interference pattern produced. Constructive interference occurs where two crests or two troughs meet while destructive interference occurs where a crest meets a trough.



The water waves are coherent: same frequency (equal to source frequency) and a constant phase difference (both dippers produce crests at the same time).

E.g. 5-11: Two identical sound sources *A* and *B* are 8.5 m apart under water and emit sound waves of frequency 3600 Hz in phase with each other. A microphone is placed on a line parallel to *AB* 750 m from the loudspeakers. If the speed of sound in water is 1530 m s^{-1} :

a) Calculate the wavelength of the sound waves in water.

b) Are the sound waves coherent? Why?

c) Describe the volume of sound at a point on the line equidistant from the speakers.

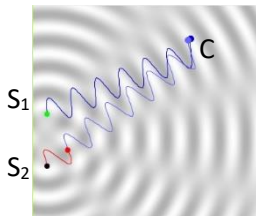
d) What happens if you move the loudspeakers closer to each other?

e) What happens if the frequency of the sound waves is increased?

f) What would happen if the waves were not coherent?

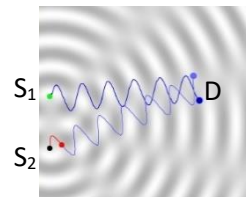
5.9.1 Interference and Path Difference

The difference in distance between two wave sources and point, presented as a multiple of the wavelength, is called the **path difference**.

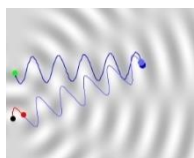


← S_1 and S_2 are two wave sources which are **in phase** and have the same wavelength. At the point C there is constructive interference because a crest from S_1 is meeting a crest from S_2 . The distance S_2C is 1 wavelength longer than S_1C , i.e. the path difference between them is 1λ .

At point D there is destructive interference because a trough from S_1 is meeting a crest from S_2 . The distance S_2D is 1 wavelength longer than S_1D , i.e. the path difference between them is $0.5\lambda \left(\frac{\lambda}{2}\right)$. →

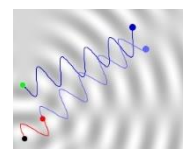


In general, for two coherent sources which are **in phase**, if the path difference is $0\lambda, 1\lambda, 2\lambda$, etc. you get constructive interference while if the path difference is $0.5\lambda, 1.5\lambda, 2.5\lambda \left(\frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots\right)$ etc, you get destructive interference.



← For two coherent wave sources in antiphase, you will get constructive interference if the path difference is $0.5\lambda, 1.5\lambda, 2.5\lambda \left(\frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots\right)$, etc

For two wave sources in antiphase, destructive interference occurs if the path difference is $0\lambda, 1\lambda, 2\lambda$, etc. →



E.g. 5-12: Identical waves, of wavelength 6 cm, are being emitted by two sources which are in phase. In each case below, state whether there is constructive or destructive interference

a) At one point, the path length difference is 12 cm.

b) There are 12 whole waves between a source and a point and 13 whole waves between the other source and the same point.

c) A point 20 cm from one wave source and 23 cm from the other:

d) A point equidistant from both sources.

E.g. 5-13: Two coherent wave sources are in antiphase. Their wavelength is 80 cm. Find if there is constructive or destructive interference at the following points:

a) At a point where the path length difference is 160 cm

b) At a point which is 170 cm from one wave source and 450 cm from the other

c) At a point equidistant from the two wave sources.

CW 5-9: **MS Forms Quiz**

HW 5-4: Interference of sound waves of wavelength 60 cm were obtained using two loudspeakers set 2.0 m apart, connected to the same signal generator.

a) Why do the loudspeakers have to be so far apart? (2)

b) Explain the term *coherence*. How is it ensured in this situation? (3)

c) You put a microphone 2.7 m from one loudspeaker and 3.3 m from the other. Find the path difference and state whether at that position you get constructive or destructive interference (2)

d) Now you put the microphone 3.2 m from one speaker and 4.1 m from the other. Find the path difference and state whether at that position you get constructive or destructive interference (2)

e) Name one way how to make the interference 'pattern' wider. (1)

5.10 Interference in Light

Trying to set up an interference pattern with light waves using two lamps is impossible because you can never get coherent light from a mixture of colours, due to all the different wavelengths. Even if we use two monochromatic lamps it will not work, even though they have the same colour - i.e. the same wavelength. Different lamps cannot maintain a constant phase relationship because the light has frequent interruptions to its smooth oscillation. These interruptions make interference between two different light sources practically impossible, since the chance of two waves having these imperfections all occurring at the same instant is virtually zero.



So, we must use a single light source, and split it in two.

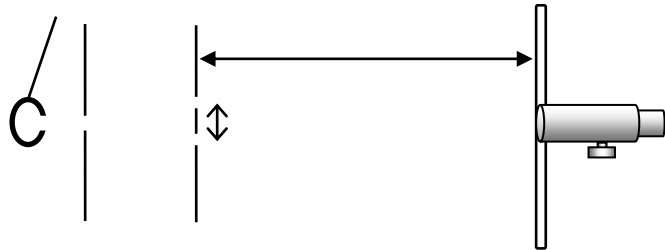
← This is what Thomas Young did! He took a beam of sunlight and split it in two, using paper.



We now do his experiment a bit differently →

5.10.1 Young's Double Slit Experiment

The apparatus (right) consists of a shielded monochromatic lamp with a narrow slit placed in front of it. The light from the narrow slit spreads out and illuminates a double slit held accurately parallel to the single slit.

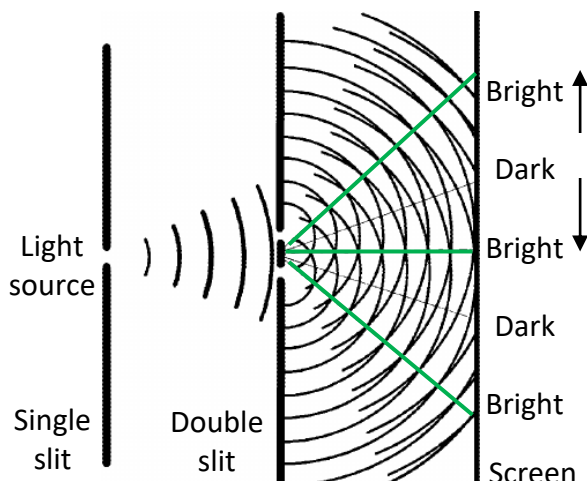


Monochromatic light is light of a single wavelength (colour)

Note: If the monochromatic lamp is replaced by a laser light, the single slit can be removed because the laser itself makes the light coherent as part of its function.

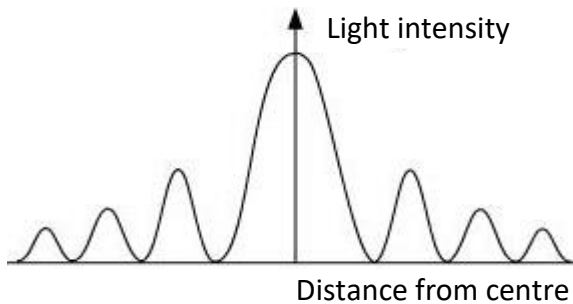
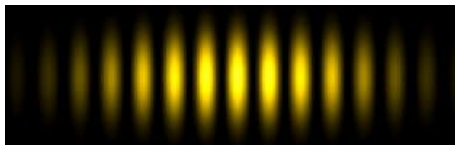
Suitable dimensions for the experiment are:

width of slits	0.2 mm
distance between double slits d	0.5 mm
distance of lamp to single slit	10 cm
distance of single slit to double slit	20 cm
distance of double slits to screen D	30 - 100 cm



The single slit ensures that the light waves falling on the double slit are in phase because the same wavefront falls on both slits. Therefore, they are coherent. This can be explained by Huygens' Principle, because the wavefronts spread out uniformly in all directions after light passes through the gap of the single slit.

The light pattern that emerges from the double slit can be viewed on a screen or with a travelling microscope.



The black lines are destructive interference, while the bright lines are constructive interference. The colour of light depends on the wavelength λ of the source.

The central fringe (called zeroth order, $n = 0$) is the brightest, i.e. it has the highest intensity. The other fringes become less bright as you move away from the centre to either side. Their order number increases in steps of 1 ($n = 1, 2, 3$ etc). The graph (left) shows how the intensity of the fringes varies with distance from the centre.

The distance between the centres of adjacent bright fringes is called the fringe separation (). It is related to wavelength, distance between double slits and distance from slits to the screen →



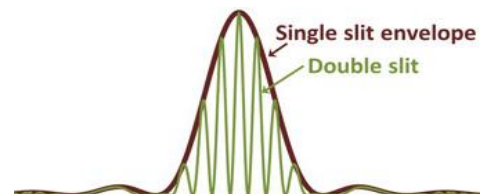
In practice, the large distance D is measured with a metre ruler while the small distances s and d measured with a travelling microscope is needed. The fringe separation y is not measured directly, but calculated by finding the average distance between many adjacent fringes, in order to reduce errors.

E.g.: Measure the distance between the centres of two fringes (not adjacent) in the above interference pattern: _____ m.

The number of spaces between the fringes is ____.

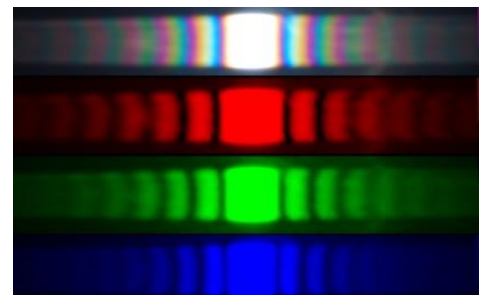
Therefore, the average fringe separation $y =$ _____

Draw a line linking the maxima of an intensity-distance graph of double slit interference to form the graph of intensity vs distance for a single slit. This line is called the single slit envelope.



5.10.2 Appearance of Young's Interference Fringes

If white light passes through the double slits, a white central fringe will be observed. All the other fringes will have coloured edges, blue being nearer to the centre because it has the shortest wavelength (see diagram below). Eventually the fringes overlap and a uniform white light is produced. This applies to passing white light through multiple slits (2 or more slits).



If the single slit is moved nearer to the double slits, the separation of the fringes is unaffected, but their brightness increases. If the single slit is widened, the fringes gradually disappear because the bright fringes overlap. If one of the slits is covered, the fringes disappear.

From the formula we can see that to increase the fringe separation y , you can either increase the distance D , use light of longer wavelength or use closer slits.

E.g. 5-14: Monochromatic light was aimed at a pair of slits 0.36 mm apart. An interference pattern forms on a screen placed 95 cm from the double slits. The distance between the bright bands was 1.2 mm. Calculate:

a) The wavelength of light used

b) The new separation between the centres of consecutive bright bands if the screen is placed 1.3 m from the slits

c) The distance between the double slits required if light of wavelength 650 nm is now used and the separation between the bands and distance between screen and slits is the same as in (b)

CW 5-10: In a Young's two-slit experiment, the bright bands of monochromatic light are 0.54 mm apart and the distance between the two slits is 1.0 mm. The distance between the slits and the focal plane of the eyepiece observing the bands is 1.20 m. Calculate the wavelength of the light. [4.50×10^{-7} m]

CW 5-11: In Young's double slit experiment, describe and explain the effect on the bands:

a) while moving the source slit nearer to the two slits

b) while decreasing the distance between the two slits

c) while moving the screen further away

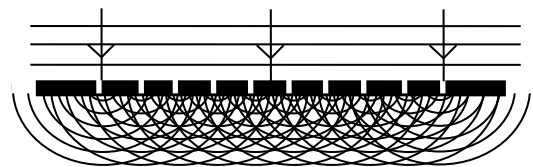
d) while replacing the monochromatic source of red light by one of blue light.

HW 5-5: To demonstrate interference of light waves, Thomas Young allowed light from an illuminated single slit to fall on a pair of narrow closely spaced slits.

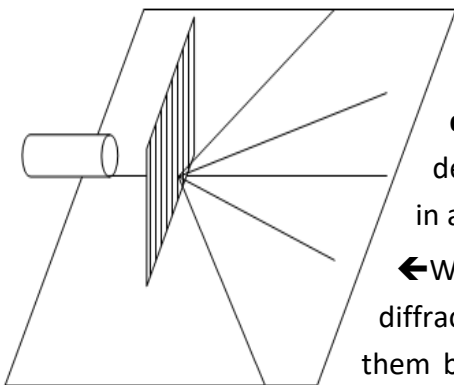
- Explain why light from two separate monochromatic lamps can never produce an interference pattern. (2)
- Explain why double slits used as described above give an interference pattern. (2)
- How would the interference pattern differ if:
 - The double slits had been closer together? (1)
 - The single slit had been much wider? (1)
 - White light had been used? (2)
- In a double slit experiment, a screen is positioned at a distance of 0.80 m from the two slits. Light of wavelength 590 nm is directed at the slits normally to give an interference pattern on the screen with three fringes spaced over a distance of 2.4 mm. Calculate the distance between the slits (2)

5.11 Diffraction Grating

In the double slit experiment, light rays from each slit interfere and produce a pattern of alternate bright and dark fringes. Using **multiple** slits, **fewer** but **sharper** bright fringes are produced, because reinforcement (i.e. constructive interference) only occurs in a limited number of directions. Each of the many slits diffracts light waves, which cancel out in most directions.



Typical diffraction gratings are 8 to 10 cm wide, with c. 6000 slits per cm. Butterfly wings & fish scales produce similar effects. German physicist von Fraunhofer made the first crude gratings.



A plate with a large number of parallel, closely spaced slits is called a **diffraction grating**. It can be used to determine the different wavelengths contained in a beam of light.

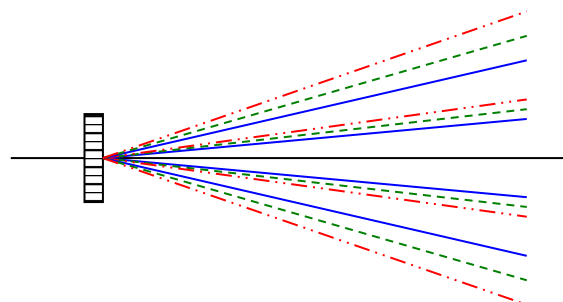


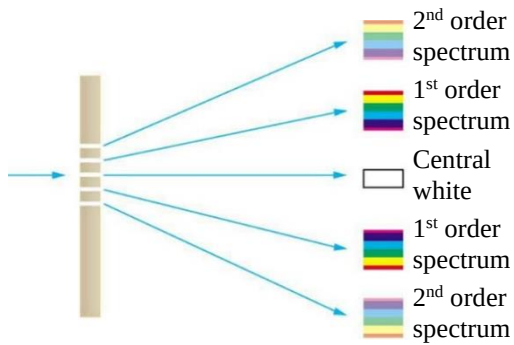
When a parallel beam of monochromatic light is incident on a diffraction grating, it is channelled in particular directions, one of them being the same as the incident light. This central direction is called the **zero order** of diffraction. The other beams are numbered 1, 2, 3 etc on both sides.



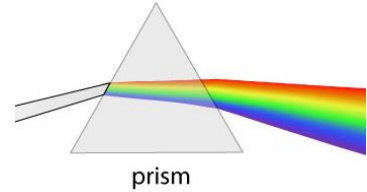
← Sharp fringes produced by a monochromatic light source

If we use mixture of wavelengths, then we will get a series of line spectra on either side of the normal (the zeroth order). Let's say that the mixture of wavelengths consists of red, green and blue light. Each order will then have these three colours in it, the shortest wavelength (blue) being closest to the central maximum.





Similarly, if white light is incident on the diffraction grating, then there will be a separation of colour, but this time we will get multiple continuous spectra on either side of the normal (the zeroth order). These continuous spectra are sharper and brighter than those produced by a prism, which produces a blurred spectrum, i.e. colours tend to merge into each other.



Let d be the distance between any two slits and let θ_n be the angle between an n^{th} order fringe and the central maximum →

The closer the slits are (i.e. the smaller d is) the more widely spaced are the diffracted beams (i.e. the bigger θ is). The angles that the diffracted light beams make with the normal of the diffraction grating are usually larger than 5° , so we cannot use $\sin \theta = \theta$.

Consider a diffraction grating with N lines (slits) per metre. Then:

The SI unit of d is m. If N is given in lines mm^{-1} , then d is in mm, which then needs to be changed to m. You can also find d from N using simple proportion.

The **greatest order** possible that one can obtain from a given diffraction grating when using a particular wavelength is found by:

Since n is a whole number, then n_{max} must be the whole number rounded down from the answer you get, because the maximum order cannot be greater than that answer.

E.g. 5-15: Monochromatic light of wavelength 600 nm is incident normally on an optical transmission grating of spacing 2.00 μm . Calculate:

a) The maximum order possible

b) The number of diffracted beams that can be observed

c) The angular positions of the maxima

E.g. 5-16: A parallel beam of monochromatic light of wavelength 650 nm is directed normally at a diffraction grating which has 600 lines per mm. Determine:

- a) The distance between any two adjacent lines.

- b) The number of diffracted beams.

- c) The angle of diffraction of the highest order beam.

E.g. 5-17: Light of wavelength 550 nm is incident normally on a diffraction grating of width 20.0 mm, on which 15000 lines have been ruled. Calculate the angular positions of the various orders.

CW 5-12: Light and sound energies both travel as waves.

- a) State two differences between a sound wave and a light wave.

- b) Explain, for each case below, why the white light is separated into its component colours, using the physical principle involved.

- i) White light is incident on a prism.

- ii) White light is incident on a diffraction grating.

c) Compare and contrast the spectrum produced by a prism and by a diffraction grating.

HW 5-6: **MS Forms Quiz**

5.12 Experiment: Diffraction Grating and Optical Spectrometer to find Wavelength

An optical spectrometer is an instrument that can be used to analyse light passing through different obstructions, like slits, prisms and diffraction gratings. It has an inbuilt protractor with a Vernier scale so that the angle θ through which light is deviated can be measured accurately to within ± 2 min of a degree.

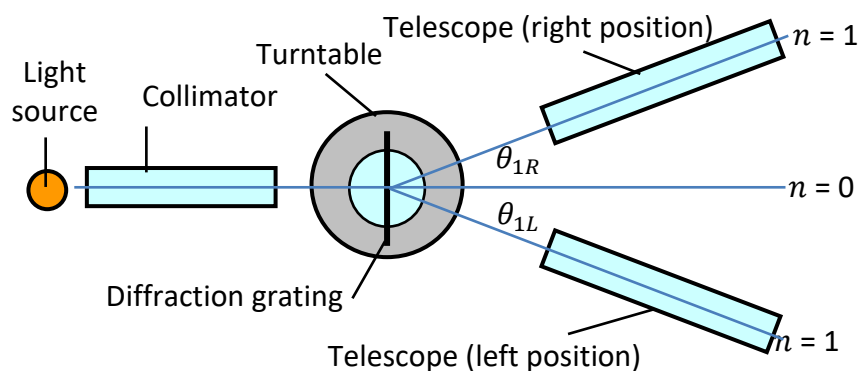


The collimator is placed in front of a light source such that the light rays coming out of it will be parallel to each other.

Aim: To measure the wavelength of a monochromatic light source.

Apparatus: Diffraction grating with a known number of lines per m, optical spectroscopy, monochromatic light source (can be a LASER, LED or another light source with a filter).

Diagram:



Method:

1. Set up apparatus such that the collimator and the telescope form a straight light.
2. Place the diffraction grating vertically in the holder.
3. Look through the eye-piece of the telescope which is straight in front of the grating to find the zero order image. Ensure the Vernier scale is zeroed.
4. Move the telescope to the left to find a sharp image of the 1st order, and measure and record the angle θ_{1L} on the protractor and Vernier scale.

5. Repeat step 4 on the right for the angle θ_{1R}
6. Find the average of θ_{1L} and θ_{1R} to obtain a value for θ_1 .
7. Repeat steps 4 to 6 for higher order images.
8. Calculate the wavelength λ for each θ_n , using the equation $\lambda = \frac{d \sin \theta}{n}$, and find their average.

Precautions:

- Make sure diffraction grating is standing vertically.
- Ensure turntable is horizontal/flat by using the levelling screws.
- Check that the Vernier callipers are set at 0° before proceeding.
- When the telescope is used to find the 0 order image, the light beam must be travelling in a straight line from the light source, through the collimator and the diffraction grating.
- A LASER ensures that the light beam is very narrow. If a LASER is not available, a narrow slit needs to be placed between the light source and the collimator to get a narrow beam.

E.g. 5-18: In an experiment to measure the wavelength of monochromatic light, a narrow beam of the light fell normally on a diffraction grating. The grating had 300 lines per millimetre. A diffraction pattern was produced. The angle between the second order image to the left and the second order image to the right of the central bright image in the pattern was measured. The angle measured was 40.6°.

- a) Draw a simple diagram of the setup used to measure the angle.

- b) Calculate the wavelength of the light used during the experiment. [5.78×10^{-7} m]

- c) Explain how using a diffraction grating of 500 lines per mm leads to a more accurate result.

CW 5-13: The following is part of a report given by a student of an experiment to measure the wavelength of monochromatic light.

"The apparatus was arranged so that bright images could be observed. The angular position θ for each of these images was determined relative to the central bright image ($n = 0$). The data obtained are shown in the table. The diffraction grating had 600 lines per mm."

$\theta / ^\circ$	n
45.0	2
20.4	1
0.0	0
20.7	1
45.2	2

- a) Use the data to calculate a value for the wavelength of the light. [5.88×10^{-7} m]

- b) Determine the maximum number of bright images that can be seen using this diffraction grating. [5]

- c) List two factors that may affect the accuracy of the experiment.

5.13 Summary of Formulae Introduced in this Topic

Description	Formula	In data booklet?
Frequency of waves on strings:	$f_n = n f_1$ $f_n = \frac{n}{2L} \sqrt{\frac{T}{\mu}}$	Y, but combined as follows: $f_n = \frac{n}{2L} \sqrt{\frac{T}{\mu}} = n f_1$
Speed of waves on strings:	$v = \sqrt{\frac{T}{\mu}}$	Y
Two slit interference:	$y = \frac{\lambda D}{d}$	Y
Diffraction grating:	$n\lambda = d \sin \theta_n$ $d = \frac{1}{N}$	Y, but written as $d \sin(\theta) = n\lambda$ Y
Single slit diffraction:	$\sin \theta = \frac{\lambda}{a}$	Y, but written as $\sin(\theta) \approx \theta = \frac{\lambda}{a}$
Diffraction of circular aperture:	$\sin \theta = \frac{1.22\lambda}{a}$	Y, but written as $\sin(\theta) \approx \theta = \frac{1.22\lambda}{a}$